Appendix A

Artificial Intelligence Use Declaration

This project was developed with assistance from AI-based development tools, consistent with the University of Hull's Student AI Use Policy (2025). This appendix provides a full account of the scope of that assistance, distinguishing clearly between AI-assisted implementation and the candidate's independent intellectual contributions.

AI-Assisted Components

AI assistance (Claude, Anthropic) was employed for the following implementation tasks:

- **Infrastructure scaffolding.** Supabase project setup, Row-Level Security policy boilerplate, Resend email integration for password reset, and bcrypt helper functions.
- **Frontend component implementation.** UI components implemented from candidate-authored specifications, including the Bloom progression indicator, XP display, and session summary layout.
- **Mapbox integration.** The `InteractiveMap` component and associated Mapbox GL JS configuration.
- **Vercel deployment configuration.** Environment variable setup, preview deployment configuration, and build pipeline scaffolding.
- **Prisma schema iteration.** Incremental schema additions from candidate-authored specifications, including migration file generation.
- **Seed script scaffolding.** Structure and helper functions for the question bank seed; all question content was authored by the candidate.

Candidate's Independent Contributions

The following constitute the candidate's independent intellectual work and were not produced by AI tools:

- The hybrid IRT 3PL + BKT algorithm architecture, including the selection of EAP over MLE estimation and the full mathematical derivation of the credit-factor BKT modification.
- The Bloom escalation gating design (0.40 and 0.60 thresholds) and its implementation as a computational selection constraint.
- The 13-KC curriculum ontology and prerequisite graph, covering UK physical geography and climate (spatial recall, climate processes, and integrative application), and its 12-edge DAG encoding causal prerequisites between knowledge components.
- The research design: within-subjects quasi-experimental structure, VanLehn benchmark selection, 7-day retention protocol, and the Sprint 5 scope reduction justification.
- The five-track multi-method evaluation framework developed in response to the $n = 4$ constraint.
- All Architecture Decision Records (ADRs) documenting algorithm selection rationale.
- All Notion sprint planning documentation, retrospective notes, and risk register entries.
- The IRT calibration strategy: fixing $c = 0.25$, monitoring RMSEA, and the two-stage R-to-Node.js parameter pipeline.

AI assistance was employed exclusively as an implementation accelerator for tasks of a non-intellectual, repetitive, or infrastructural nature. All design decisions, algorithmic contributions, and evaluation methodology are the candidate's own work and are grounded in the peer-reviewed literature cited throughout this report.